Extraction games with a priced adversary

An intermediate report for the survey's formalization

2026-08-22, second pass

1 What this report is

This report turns a set of proposed games into definitions, states what each one settles and what it does not, and identifies the reduction targets available to them. It is written to be reused. Sections 4 to 6 are candidate text for the survey's formalization section, and Section 9 is the part that must be settled before any of it carries a theorem.

Three commitments hold throughout. Every game names a baseline, because a success probability without one is not a measurement. Every game prices both parties, because the object of interest is a ratio. And every claim is marked as proved here, assumed, or open.

Scope. The games below assume a metric output space in which a tolerance ε is meaningful, so they describe regression and real-valued-output models. Under a 0/1 output metric, Propositions 1 and 3 are vacuous, because no useful Lipschitz constant exists. The classification case needs a different treatment and this report does not give one.

2 Notation

Fix an input space $\mathcal{X}$ with metric $d_{\mathcal{X}}$ and an output space $\mathcal{Y}$ with metric $d_{\mathcal{Y}}$. We use the pipeline decomposition of Villa et al. [24], in which a system is a tuple $(\textsc{GenerateData}, \textsc{Learn}, \textsc{Infer})$ carrying a source to a corpus, a corpus to a model, and a model with an input to a result. We write ATK for their capability set, with elements $\mathsf{see_model}$, $\mathsf{black_box}$, $\mathsf{inject_learn}$ and $\mathsf{inject_infer}$.

Everything is indexed by a security parameter n, and this is not a formality. Without it the hardness statement in Section 6 cannot be written. Let $\mathcal{M}_n$ be a distribution over models $f : \mathcal{X}_n \to \mathcal{Y}_n$, induced by running $(\textsc{GenerateData}, \textsc{Learn})$ on a source. The learner $\mathcal{L}$ holds $f \sim \mathcal{M}_n$. We write $C_{\textsc{Infer}}$ for the cost of one honest evaluation of f, and measure every party's budget as a dimensionless multiple of it.

3 The extrapolation game with an adversary-chosen probe

The first proposal is the most natural. The adversary queries, then names a point it did not query and predicts the model's value there.

Game 1 ($\mathsf{Extr}^{\mathrm{adv}}$)**.** Parameters: a query budget m, a probe radius Δ, a tolerance ε.

1. $\mathcal{A}$ chooses $Q = \{q_1, \ldots, q_m\} \subset \mathcal{X}$ and receives $R = \{r_i = f(q_i)\}_{i \le m}$.
2. $\mathcal{A}$ outputs (q', r') with $q' \notin Q$ and $d_{\mathcal{X}}(q', Q) \le \Delta$.

3. $\mathcal{A}$ wins if $d_{\mathcal{Y}}(r', f(q')) \le \varepsilon$.

This game is not usable as stated, and the reason constrains every later version.

Proposition 1 (The probe collapses)**.** *Let f be L-Lipschitz from $(\mathcal{X}, d_{\mathcal{X}})$ to $(\mathcal{Y}, d_{\mathcal{Y}})$. Suppose the rules of Game 1 admit some q' with $0 < d_{\mathcal{X}}(q', q_i) \le \delta$ for a queried q_i, and that $L\delta \le \varepsilon$. Then the specific adversary that queries one point q_i, names such a q' and outputs $r' = r_i$ wins with probability* 1*, using one query and no computation beyond naming q'.*

Proof. $d_{\mathcal{Y}}(r_i, f(q')) = d_{\mathcal{Y}}(f(q_i), f(q')) \le L\, d_{\mathcal{X}}(q_i, q') \le L\delta \le \varepsilon$. □

Proposition 1 is an existence statement about one adversary, and that is enough to reject the game. Where such a q' is admissible the game measures the modulus of continuity of f rather than the adversary. Two repairs exist. The first forces the probe away from the queried set, requiring $d_{\mathcal{X}}(q', Q) \ge \delta_{\min}$ with $L\delta_{\min} > \varepsilon$, which makes the difficulty depend on L, a property of the model that the adversary may not know and that the isoperimetry line studies in its own right [8]. The second removes the adversary's choice, and is the one developed here.

4 The learner-chosen probe

Let the learner name the challenge points. The geometry disappears, because success is measured where the learner chose rather than where the adversary argued was close enough.

Game 2 ($\mathsf{Extr}^{\mathrm{lrn}}$)**.** Parameters: a query budget m, a challenge count k, a tolerance ε, a challenge distribution $\mathcal{Q}$ over k-subsets of $\mathcal{X}$, a mode $\mathsf{mode} \in \{\mathsf{commit}, \mathsf{adapt}\}$. Sample $f \sim \mathcal{M}_n$.

1. If $\mathsf{mode} = \mathsf{commit}$, $\mathcal{L}$ samples $Q^* \sim \mathcal{Q}$ and publishes a commitment to it.
2. $\mathcal{A}$ chooses $Q = \{q_1, \ldots, q_m\}$ and receives $R = \{f(q_i)\}_{i \le m}$.
3. If $\mathsf{mode} = \mathsf{adapt}$, $\mathcal{L}$ samples $Q^* \sim \mathcal{Q}(Q)$ now. $\mathcal{L}$ sends $Q^* = \{q_1^*, \ldots, q_k^*\}$, opening the commitment in the committed mode.
4. $\mathcal{A}$ outputs $(r_1^*, \ldots, r_k^*)$.
5. $\mathcal{A}$ wins if $d_{\mathcal{Y}}(r_j^*, f(q_j^*)) \le \varepsilon$ for every $j \le k$.

The two modes differ in what they measure. In the committed mode the challenge is independent of the adversary's queries, so the game measures generalization from a fixed test set. In the adaptive mode the learner may place challenges where the adversary did not look, so the game measures coverage. That the adaptive mode is harder for the adversary is intuitive and is not proved here; a coupling argument would settle it, and Open Problem 2 records it.

4.1 The advantage, and why the baseline is part of the definition

The baseline must be defined so that it cannot smuggle in the model. If f is fixed and the baseline maximizes over all algorithms that "do not access f", the maximizing algorithm can simply have f hardwired, the baseline equals the honest success probability, and the advantage is non-positive for every adversary. That is the same degeneracy that makes the marginal-advantage form of membership advantage a no-op. The repair is to draw the model from a distribution and measure the baseline against that distribution, which is what Balle, Cherubin and Hayes do by drawing the target record from a prior [4].

Definition 1 (Extraction advantage)**.** Let $\mathcal{B}$ range over algorithms that receive the public description of $\mathcal{M}_n$, the task, and the challenge set Q^*, but neither f nor the transcript (Q, R). Then

$$\mathsf{Adv}^{\mathsf{Extr}}_{\mathcal{M}_n,\mathcal{Q}}(\mathcal{A}) = \Pr\left[\mathcal{A} \text{ wins}\right] - \max_{\mathcal{B}} \Pr\left[\mathcal{B} \text{ wins}\right],$$

both probabilities over $f \sim \mathcal{M}_n$, over $\mathcal{Q}$, over the coins of $\mathcal{A}$ and $\mathcal{B}$, and over any randomness in Infer.

Because f is random and $\mathcal{B}$ is quantified before f is drawn, no $\mathcal{B}$ can hardwire f, and the baseline is the best model-free predictor for the *family* rather than for one model. Definition 1 still differs from reconstruction robustness in exactly one place, and the difference is the point of this report: Balle, Cherubin and Hayes place no bound on the adversary's computation, quantifying over any reconstruction map, while Section 6 adds one. Their R is the attack, and the criterion is a separate error function with tolerance η.

Proposition 2 (The raw success probability is not a measurement)**.** *There exist a task, a model family and a challenge distribution for which a model-free $\mathcal{B}$ attains success probability strictly greater than that of a published attack with full query access.*

Status and evidence. Empirical, and established for membership inference rather than for extraction. Das et al. exhibit it on eight published membership-inference benchmarks, where classifiers with no model access outperform the best reported attack that queries the model [15]. Meeus et al. report the same effect and identify the carrier, a distribution shift between the two halves of each benchmark [21]. Whether the same construction arises for extraction challenge distributions is asserted here and not exhibited. □

4.2 Separated challenges

The extension to k challenges is meant to measure utility rather than one lucky point. Without a constraint on how the challenges sit relative to one another it measures neither.

Proposition 3 (Unseparated challenges collapse to one)**.** *Let f be L-Lipschitz. Suppose the support of $\mathcal{Q}$ is contained in a ball $B(c, \rho)$ whose centre c is known to the adversary before it queries, and suppose $L\rho \le \varepsilon$. Then the adversary that queries c once and answers every challenge with $f(c)$ wins Game 2 with probability* 1*, for every k, in either mode.*

Proof. For each j, $d_{\mathcal{Y}}(f(c), f(q_j^*)) \le L\, d_{\mathcal{X}}(c, q_j^*) \le L\rho \le \varepsilon$, and the win condition is a conjunction over j. □

The hypothesis that the ball is fixed and known before step 2 is necessary, because the adversary queries before Q^* is revealed. Two constraints repair the game, and they are not equivalent.

Definition 2 (Δ-separated challenge distribution)**.** $\mathcal{Q}$ is Δ-separated if every Q^* in its support satisfies $\min_{j \ne j'} d_{\mathcal{X}}(q_j^*, q_{j'}^*) \ge \Delta$. The Δ-packing number of $\mathcal{X}$ upper bounds the challenge count k.

Definition 3 (h-unpredictable challenge distribution)**.** $\mathcal{Q}$ is h-unpredictable if $H_\infty(Q^*) \ge h$, so no algorithm guesses Q^* with probability better than 2^{-h}.

Definition 2 is geometric and rules out Proposition 3. Definition 3 is information-theoretic and rules out an adversary that precomputes answers for a predictable challenge set. A game that prices an adversary needs both, because packing alone permits a fixed challenge set and min-entropy alone permits a clustered one. The pair is the analogue of requiring distinct and unpredictable challenge messages in an indistinguishability game.

5 The decision variant

The search game asks the adversary to produce a value. The crypto-native form asks it to distinguish, giving an advantage against a coin rather than against a maximization over baselines.

Game 3 (Extr-IND)**.** As Game 2 through step 3. Then $\mathcal{L}$ samples $b \leftarrow \{0,1\}$. If $b = 0$ it returns $y_j = f(q_j^*)$ for each j. If $b = 1$ it returns $y_j \sim \mathcal{R}(q_j^*)$, independently across j, for a reference distribution $\mathcal{R}$ independent of f. $\mathcal{A}$ outputs b' and wins if $b' = b$. Its advantage is $\mathsf{Adv}^{\mathsf{IND}}(\mathcal{A}) = |\Pr[b' = b] - \frac{1}{2}|$.

Proposition 4 (Search implies decision)**.** *Let $\mathcal{A}$ win Game 2 with probability p. Define*

$$q \;=\; \mathbb{E}_{f,Q^*,\text{coins of } \mathcal{A}}\left[\prod_{j\le k} \Pr_{y_j\sim\mathcal{R}(q_j^*)}\left[\, d_{\mathcal{Y}}(y_j, r_j^*) \le \varepsilon \,\right]\right],$$

the probability that a full reference vector falls within tolerance of $\mathcal{A}$'s own predictions. Then there is a distinguisher $\mathcal{A}'$ with $\mathsf{Adv}^{\mathsf{IND}}(\mathcal{A}') = \frac{1}{2}|p-q|$, running in the time of $\mathcal{A}$ plus k distance evaluations.

Proof. $\mathcal{A}'$ runs $\mathcal{A}$ to completion on (Q^*, R) to obtain $(r_1^*, \dots, r_k^*)$, and only then reads $(y_1, \dots, y_k)$. It outputs $b' = 0$ if $d_{\mathcal{Y}}(y_j, r_j^*) \le \varepsilon$ for all j, and $b' = 1$ otherwise. Because $\mathcal{A}$ has finished before y is read, the vector r^* is independent of y given Q^*, and because the y_j are independent across j the acceptance probability under $b = 1$ is the product above. Hence $\Pr[b' = 0 \mid b = 0] = p$ and $\Pr[b' = 0 \mid b = 1] = q$, so $\Pr[b' = b] = \frac{1}{2}(p + 1 - q)$ and the advantage is $\frac{1}{2}|p - q|$. □

The converse is the interesting direction and is not available in general. The nearest published statement is Alvarez-Melis, Garg and Kalai [1], and it must be quoted carefully because the paper describes the same reduction from both ends. Its introduction calls the result "a polynomial-time reduction from likelihood maximization to next-token distinguishability", while the body concludes that its Algorithm 1 with Theorem 1 and Lemma 3 "yields a polynomial-time reduction from distinguishing distributions to maximum likelihood estimation for sequential models". The unambiguous operational statement is that Algorithm 1 converts a next-token distinguisher into a likelihood improvement, so a distinguisher yields a better generator, and no converse is stated in that model. Two further boundaries hold. The two-directional result is in a different section and is not polynomial time, the paper noting that repeating its efficient step gives an exponential-time algorithm overall. And no distinguisher anywhere in the paper carries a runtime bound, although the next-token result does fix a distinguisher class supplied by an oracle.

Open problem 1. *Is there a polynomial-time reduction from* Extr-IND *to* $\mathsf{Extr}^{\mathrm{lrn}}$ *under a Δ-separated, h-unpredictable challenge distribution? A positive answer makes the decision game the definition of record and the search game a corollary.*

Open problem 2. *Prove by coupling that the adaptive mode of Game 2 is at least as hard for the adversary as the committed mode, or exhibit a $\mathcal{Q}$ where it is not.*

6 Pricing the two parties

Every game above is a predicate on a transcript. It becomes a security notion only when both parties carry a budget, and the budget must be written so that the adversary cannot inflate its own allowance.

Measure cost in evaluations of f, so all quantities are dimensionless. The adversary's total cost is $t_{\mathcal{A}}(n) = m + T_{\mathcal{A}}/C_{\textsc{Infer}}$, where m is its query count and $T_{\mathcal{A}}$ its own computation. The honest reference cost is $k(n)$, the number of evaluations the learner needs to produce the k challenge answers it is adjudicating against.

The reference cost deliberately excludes both the m queries the learner answers and the cost of training f. Excluding the queries matters, because a threshold that depended on m would rise as the adversary queried more, letting the adversary raise its own admission bar. Excluding training matters because a threshold dominated by pre-training is met by no query-bounded adversary and the definition would be vacuous.

Definition 4 (δ-extraction hardness)**.** A family $\{(\mathcal{M}_n, \mathcal{Q}_n, k(n), \varepsilon(n))\}_n$ is δ-extraction hard if for every adversary $\mathcal{A}$ with $t_{\mathcal{A}}(n) < k(n)^{1+\delta}$ for all large n,

$$\mathsf{Adv}^{\mathsf{Extr}}_{\mathcal{M}_n,\mathcal{Q}_n}(\mathcal{A}) \;\le\; \mathrm{negl}(n).$$

The name carries only δ, deliberately. Calling it (ε, δ)-hardness would collide with Brzuska and Couteau [7], where ε bounds the inverter's success probability, whereas ε here is the game's tolerance. Three cautions apply and the survey must state all three.

1. **Three conventions, not one.** Brzuska and Couteau use non-uniform circuit size with the honest cost raised to $1 + \delta$ [7]. LaVigne, Lincoln and Vassilevska Williams use uniform word-RAM time and name the gap as a constant δ separating honest $T^{1-\delta}$ from the threshold T [20]. Ball, Rosen, Sabin and Vasudevan [2] index their error by the adversary's own discount as well as by n, so one number does not describe their notion. Quoting one convention's δ in another's units is an error, and Definition 4 is in the first convention.

2. **The known ceiling is a query bound.** Barak and Mahmoody [5] show an adversary making $O((n/\delta)^2)$ *oracle queries* breaks any n-query key exchange in the random oracle model, and note their adversary runs efficiently in a relativized world where P = NP. That forbids a super-quadratic *query* gap in an idealized model. It says nothing about time gaps or structured assumptions, so Definition 4 is not capped at $\delta = 1$ by it.

3. **The exponent is on a dimensionless count.** Raising a cost in seconds or FLOPs to the power $1 + \delta$ is not invariant under a change of unit. Measuring in evaluations of f removes the ambiguity, and matches Brzuska and Couteau, whose $|f|$ is a circuit size.

7 Model stealing under an abstracted distance

Extraction of the function itself is the same game with the output replaced by a model. The difficulty is that the natural distance is the wrong one. Two networks that agree everywhere differ in weights, and a transformer's trunk is largely a product of pre-training that no attacker needs to steal.

Let ρ be a *read-out map* carrying a model to the part of it that matters, and let $D_\rho(f, f') = d(\rho(f), \rho(f'))$.

Game 4 (Steal_ρ)**.** As Game 2 through step 2. Then $\mathcal{A}$ outputs a model f', and wins if $D_\rho(f, f') \le \varepsilon$.

Proposition 5 (Stealing implies answering)**.** *Let ρ restrict a model to the support of $\mathcal{Q}$ and let d be the supremum distance. If $\mathcal{A}$ wins Steal_ρ with tolerance ε, then evaluating f' at each challenge wins Game 2 with the same ε, at the cost of k evaluations of f'.*

Proof. $d_{\mathcal{Y}}(f'(q_j^*), f(q_j^*)) \le \sup_{q \in \operatorname{supp} \mathcal{Q}} d_{\mathcal{Y}}(f'(q), f(q)) = D_\rho(f, f') \le \varepsilon$ for every j. □

The converse fails without a covering condition. Answering k separated challenges pins f at k points, and k points determine f on the support only if they form an α-net for a modulus of continuity bounding f. This is where Definition 2 does work beyond ruling out Proposition 3: the packing number bounds how many separated challenges exist, and the covering number bounds what k challenges can certify.

Open problem 3. *For which $(\rho, \mathcal{Q}, k)$ does winning Game 2 on a Δ-separated challenge set imply winning Steal_ρ? A quantitative answer converts a query-complexity bound into a model-extraction bound and back.*

The choice of ρ decides what the theorem is about and must be stated rather than assumed. Jagielski et al. [18] separate the two natural choices as accuracy and fidelity objectives, and show they conflict for an imperfect victim, since a high-fidelity copy must reproduce the victim's errors and a high-accuracy copy must not. They also dismiss exact weight recovery as the wrong target, because ReLU networks sit in large behavioural equivalence classes closed under neuron rescaling, permutation and dead-neuron insertion. Any ρ used here must be invariant under those symmetries or it measures an artifact.

8 The challenge generator as an oracle

In the adaptive mode the learner chooses Q^* after seeing the adversary's queries. That choice is an algorithm, and leaving it unspecified hides a party whose power decides the game.

Model it as an auxiliary oracle $\mathcal{O}_{\text{chal}}$ in the sense of Villa et al. [24], with its own (GenerateData, Learn, Infer and capability set. It need not be a language model. A derivative of f, or a smaller model trained on the same task, suffices.

Three consequences follow.

1. The capability set becomes the access coordinate of the survey's frame directly. black_box and see_model separate the query and white-box variants of every game above, and inject_learn opens the poisoning variant in which $\mathcal{A}$ influences f before the game.

2. Villa et al. separate completeness, measured with no adversary, from security, measured against one. Game 2 needs the same split, because a challenge set the honest learner cannot itself answer within ε makes the game unwinnable for reasons unrelated to the adversary.

3. A filtering impossibility bears on how cheap $\mathcal{O}_{\text{chal}}$ can be, and its attribution matters. The result is Ball et al. [3], which Villa et al. discuss rather than prove. It is *existential*: the abstract states that there exist language models for which no efficient prompt filter works, and the construction assumes the model shares a secret key with the attacker. It does not say that every judge must match the complexity of its predicate. Read at its actual strength it suggests, and does not establish, that a challenge generator too weak to compute the property it tests cannot place hard challenges, which would put $\mathcal{O}_{\text{chal}}$'s own cost inside the honest budget.

9 Where to reduce

The proposal asks what to reduce to when there is no instance in hand. Four answers exist.

A definitional target with no reduction. A game may stand as a definition before any instantiation. What this buys is a statement other results can be mapped into, which is the survey's goal. What it costs is that no theorem follows. The programme's own standing constraint is that the machinery here is off-the-shelf, so the contribution must be a theorem at a stated δ and not a definition. This option therefore satisfies the survey and not the thesis, and saying so is part of the report.

Learning-theoretic exponents. Two distinct papers are involved and conflating them is easy. Damian, Lee and Bruna [12] give $n = \Theta(d^{1 \vee k^*/2})$ for Gaussian multi-index models with the generative leap exponent, with a low-degree lower bound and a matching efficient upper bound whose constant $C(P, K)$ is conjectured exponentially large in the index dimension. The detection caveat belongs to the single-index paper, Damian, Pillaud-Vivien, Lee and Bruna [13], whose Remark 3.3 says its theorem primarily measures a detection threshold, and whose low-degree half is conditional on its Conjecture 3.4. An adversary that must recover a value is doing estimation, whose threshold those authors leave open. Separately, multi-pass gradient descent with batch reuse escapes the information and leap exponents [14], so these price single-pass training rather than intrinsic difficulty.

Planted problems. Planted clique, planted subgraph and continuous LWE supply average-case hardness with a long record. The caution is that restricted-class evidence and the security statement are different objects. Bogdanov, Jones, Rosen and Zadik [6] build private simultaneous messages and secret sharing whose security quantifies over size-bounded circuits, with low-degree analysis offered as evidence for the underlying conjectures. No security definition there quantifies over a low-degree adversary. Buhai, Hsieh, Jain and Kothari [9] disprove the quasi-polynomial low-degree conjecture in the regime it was stated, so low-degree-bounded and quasi-polynomial-bounded are provably different. Two clauses travel with that: their polynomial-time distinguisher is for the rectangular generalisation, while in the square symmetric regime it is $n^{O(\log^{1/(k-1)} n)}$, and the positive counterpart [17] must be cited alongside.

Fine-grained hard problems. Average-case k-SUM and Zero-k-Clique carry fine-grained hardness and are the established route to a statement shaped like Definition 4 [20, 2]. This is the target that matches the quantity, because the conclusion wanted here is a ratio and these are the only assumptions stated as ratios. The cost is a theorem about a constructed family rather than a deployed model, a boundary the survey already commits to stating.

10 What is already occupied

- **The criterion, tolerance and baseline exist.** Balle, Cherubin and Hayes [4] define reconstruction robustness with an error function, a tolerance η, and a baseline given by the best attack that ignores the model. What is missing there is any bound on the adversary's computation.
- **An extraction adversary priced by algorithm class exists, in a security paper.** Jagielski et al. [18] state as their Theorem 3, imported from Das et al., that model extraction approaches expressible as statistical-query algorithms require $\exp(O(h))$ samples for fidelity extraction of depth-h random networks, and note that gradient descent and PCA are such

algorithms. This is closer to the present proposal than anything else found, and any claim of novelty must be stated against it.

- **Charging the adversary compute per query is deployed.** Dziedzic et al. [16] wrap every prediction in a proof-of-work puzzle whose difficulty is calibrated to how much information the client has already extracted, reporting under $2\times$ slowdown for legitimate users and up to $100\times$ for attackers. It carries no definition and no theorem, which is the gap, but "prices the attack in compute" without that qualifier is already published.

- **The query budget is already a definitional parameter.** Chandrasekaran et al. [11] give the pre-2023 literature its closest thing to a formal extraction definition, with the query budget inside it. Their reduction turns a query-synthesis active learner into an extraction adversary, so extraction reduces to active learning and the complexity figures travel in the upper-bound direction only. The converse, that a class hard to actively learn is hard to extract, is neither proved nor claimed.

- **Extraction as cryptanalysis exists.** Carlini, Jagielski and Mironov [10] recover ReLU parameters by differential cryptanalysis from real-valued oracle access. The attack uses polynomially many queries but work exponential in layer width for sign recovery, and it is empirical.

- **The success definition is reported missing from inside.** Shafran et al. [22] state that they are aware of no model-extraction literature that formally defines what makes an attack successful. The scope of that statement is accuracy-driven extraction, with fidelity and functionally equivalent extraction excluded, and the work is a preprint.

- **The attacker-versus-honest ratio is stated, and parameterized by a constant but never by an exponent.** Proof-of-learning requires that forging cost at least as much as honest training, in its Property 2 and its Definition 2, offering a conjecture in place of a proof [19]. Its Property 1 is a different inequality, bounding verification against training. Later work does index the ratio, by a multiplicative constant in Fang et al. [25] and by efficiency and hardness parameters in the compute-monitoring proposal of Shavit [23]. What no work does is index it by an exponent, which is the only form Definition 4 adds.

11 Open problems, in the order they block progress

1. Open Problem 1, the polynomial-time converse of Proposition 4.
2. The covering condition under which Game 2 implies Steal_ρ, quantitatively in k and Δ.
3. A read-out map ρ invariant under the ReLU symmetries that Jagielski et al. name, so that D_ρ measures function and not parameterization.
4. An instance meeting Definition 4 at an explicit δ, reduced to average-case k-SUM or Zero-k-Clique.
5. A separation between Definition 4 and the statistical-query bound of Jagielski et al. Theorem 3, exhibiting a family hard for a compute-bounded adversary and easy for an unbounded statistical-query one, or the reverse. Without it the exponent form adds no theorem.
6. The classification case, where the tolerance ε and the Lipschitz arguments of Section 3 do not apply.

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